Aging Estimation of an IGBT-Based Drive

Arrhenius model with joint current and coolant temperature thresholds

August 5, 2026

1 Scope and assumptions

This document defines an empirical model for the cumulative aging estimation of a liquid-cooled drive, based on quantities available online:

- delivered current I (phase RMS value);
- heatsink inlet water temperature T_w;
- IGBT junction temperature T_j (measured or estimated);
- number of maintenance or repair interventions N performed on the drive;
- IGBT module type, $\tau \in \{1, 2\}$.

The approach follows the physics-of-failure framework commonly used for reliability assessment of power converters [3, 4, 2].

Base assumptions:

1. the dominant degradation mechanism is thermally activated, hence describable by an Arrhenius law on junction temperature;
2. current acts as a multiplicative stress factor (losses, current density in the bond wires, thermal cycle amplitude);
3. an *out-of-design* operating condition exists when the coolant temperature exceeds a threshold *simultaneously* with the current exceeding a threshold; in that condition the aging rate is further accelerated;
4. the type 1 module has an aging coefficient twice that of the type 2 module;
5. each maintenance intervention leaves a residual penalty (reassembly, thermomechanical stress on the power stack, degradation of the thermal interface).

2 Notation

3 Model overview

Figure 1 summarises the model structure: measured quantities on the left, acceleration factors in the centre, damage accumulator and diagnostic outputs on the right.

Figure 2 shows the operating plane (I, T_w) with the critical quadrant where the additional acceleration factor is activated: activation requires the two thresholds to be exceeded *simultaneously*.

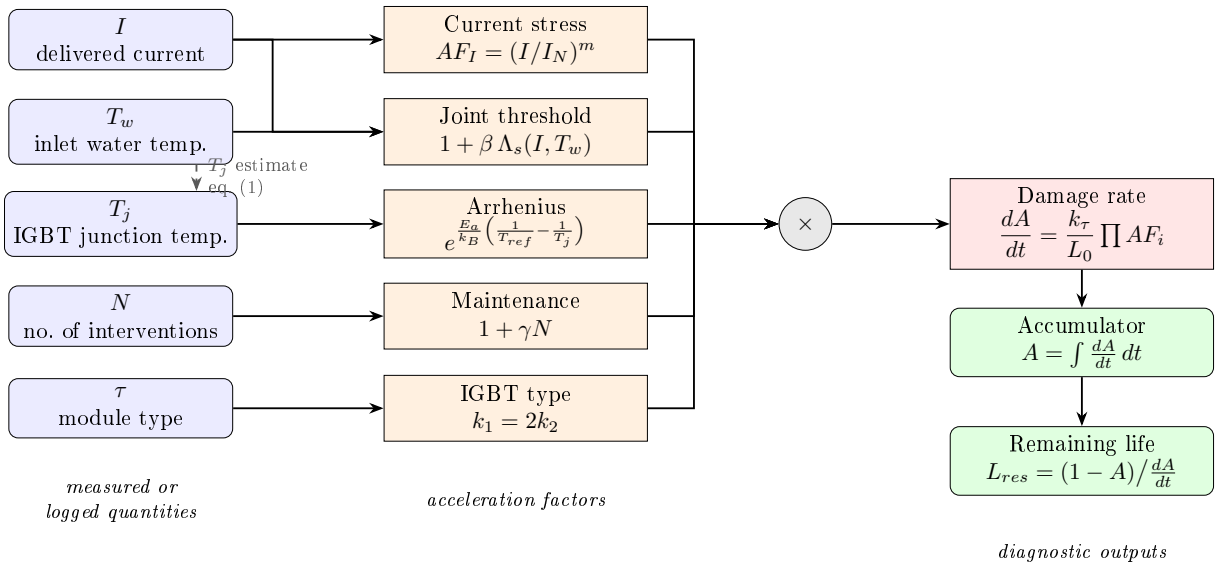


Figure 1: Structure of the aging estimation model: measured inputs generate multiplicative acceleration factors which, combined with the module type coefficient k_τ, define the life consumption rate; time integration yields the cumulative index and the remaining life.

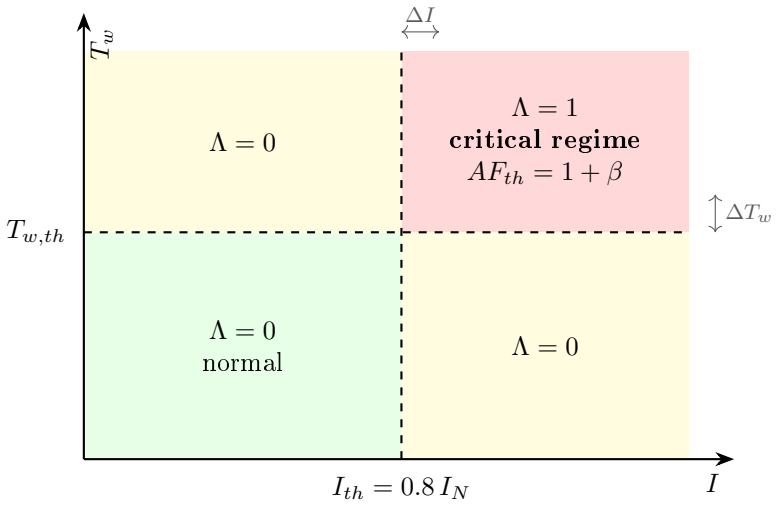


Figure 2: Current versus water temperature operating plane. The additional acceleration factor is activated only in the upper-right quadrant; ΔI and ΔT_w are the transition widths of the regularised formulation (5).

Symbol	Meaning	Unit
I	delivered current (RMS)	A
I_N	rated drive current	A
I_{th}	current threshold for the accelerated regime	A
T_w	inlet water temperature	°C
$T_{w,th}$	water temperature threshold	°C
T_j	IGBT junction temperature	K
T_{ref}	model reference temperature	K
E_a	activation energy	eV
k_B	Boltzmann constant, 8.617×10^{-5}	eV/K
m	current stress exponent	–
β	acceleration gain in the threshold regime	–
γ	penalty per maintenance intervention	–
k_τ	aging coefficient of the module type	–
L_0	expected life at reference conditions	h
A	cumulative aging index, $A \in [0, 1]$	–

Table 1: Symbols used in the document.

4 Simplified thermal model

If T_j is not available from direct measurement (module NTC, $V_{ce,on}$ used as a thermo-sensitive electrical parameter [8]), it can be estimated in quasi-steady-state conditions from T_w and the losses:

$$T_j = T_w + R_{th,jw}\, P_{loss}(I), \qquad P_{loss}(I) \simeq a_1 I + a_2 I^2, \tag{1}$$

where the linear term represents the switching losses (approximated as proportional to the switched current) and the quadratic term the conduction losses across the differential resistance. $R_{th,jw}$ includes junction-to-case, thermal interface and case-to-water contributions, and is a function of the coolant flow rate.

5 Arrhenius acceleration factor

The acceleration factor with respect to the reference temperature is

$$AF_{Arr}(T_j) = \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{ref}} - \frac{1}{T_j}\right)\right]. \tag{2}$$

For $T_j > T_{ref}$ we get $AF_{Arr} > 1$ (accelerated aging), for $T_j < T_{ref}$ we get $AF_{Arr} < 1$. Temperatures must be expressed in kelvin. Typical E_a values for power modules range between 0.4 eV and 0.8 eV depending on the dominant mechanism (metallisation degradation, electromigration, gel degradation) [1, 14].

6 Current stress factor

$$AF_I(I) = \left(\frac{I}{I_N}\right)^m, \qquad m \in [1.5,\, 4]. \tag{3}$$

The exponent m captures, in compact form, the dependence of damage on thermal cycle amplitude (Coffin–Manson type, $N_f \propto \Delta T_j^{-n}$, with ΔT_j increasing with I). Reference power cycling lifetime models for IGBT modules are LESIT [5], CIPS08 [6] and Scheuermann [7].

7 Joint threshold condition

The accelerated aging regime is activated when *both* conditions hold:

$$\Lambda(I, T_w) = \mathbb{1}[T_w > T_{w,th}] \cdot \mathbb{1}[I > I_{th}] \in \{0, 1\}. \tag{4}$$

The step formulation (4) is discontinuous and, in the presence of measurement noise, causes repeated toggling of the index. A regularised version based on sigmoid functions is therefore preferred:

$$\Lambda_s(I, T_w) = \sigma\left(\frac{T_w - T_{w,th}}{\Delta T_w}\right) \cdot \sigma\left(\frac{I - I_{th}}{\Delta I}\right), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}, \tag{5}$$

with ΔT_w and ΔI transition widths (for example 2 K and $0.05\, I_N$). Alternatively, the step form can be kept by adding hysteresis on both thresholds.

The overall acceleration factor of the out-of-design condition is

$$AF_{th}(I, T_w) = 1 + \beta\, \Lambda_s(I, T_w), \tag{6}$$

where $\beta = 1$ corresponds to doubling the aging rate in the critical regime.

8 IGBT type

$$k_\tau = \begin{cases} 2\, k_0 & \tau = 1 \quad \text{(type 1 module)} \\ k_0 & \tau = 2 \quad \text{(type 2 module)} \end{cases} \qquad k_0 = 1. \tag{7}$$

For the same mission profile, the type 1 module consumes life at twice the rate of the type 2 module.

9 Maintenance interventions

Each intervention on the drive introduces a permanent penalty, modelled as a linear multiplicative factor:

$$AF_N(N) = 1 + \gamma N, \qquad \gamma \in [0.02,\, 0.10]. \tag{8}$$

The counter N is incremented at each logged intervention (module replacement, opening of the cooling circuit, reassembly of the power stack). To distinguish intervention severity, γN can be replaced by $\sum_{i=1}^{N} \gamma_i$, with γ_i weighted by intervention class.

10 Complete formulation

The instantaneous life consumption rate is

$$\boxed{\frac{dA}{dt} = \frac{k_\tau}{L_0} \left(\frac{I}{I_N}\right)^m \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{ref}} - \frac{1}{T_j}\right)\right] \left[1 + \beta\, \Lambda_s(I, T_w)\right] \left(1 + \gamma N\right)} \tag{9}$$

and the cumulative aging index

$$A(t) = \int_0^t \frac{dA}{d\xi}\, d\xi, \qquad A \in [0, 1], \tag{10}$$

with $A = 0$ for a new drive and $A = 1$ at conventional end of life. The remaining life estimated at constant operating conditions is

$$L_{res} = \frac{1 - A(t)}{\dfrac{dA}{dt}}. \tag{11}$$

11 Discrete-time implementation

With sampling period T_s (typically 1 s to 60 s, since thermal dynamics are slow compared to the control loop):

$$A[k+1] = A[k] + T_s \, \frac{k_\tau}{L_0} \left(\frac{I[k]}{I_N} \right)^m AF_{Arr}\big(T_j[k]\big) \big[1 + \beta \Lambda_s\big] \big(1 + \gamma N[k]\big). \tag{12}$$

Listing 1: Pseudo-code for embedded implementation.

```
/* executed every Ts seconds */
Tj_K  = Tj_C + 273.15f;
af_arr = expf( (Ea/kB) * (1.0f/Tref_K - 1.0f/Tj_K) );
af_i   = powf( I/IN, m );
lam    = sigm((Tw - Tw_th)/dTw) * sigm((I - I_th)/dI);
af_th  = 1.0f + beta*lam;
af_n   = 1.0f + gamma*(float)N;
k_tau  = (igbt_type == 1) ? 2.0f : 1.0f;

rate   = (k_tau/L0) * af_i * af_arr * af_th * af_n;   /* 1/h */
A     += rate * (Ts/3600.0f);
if (A > 1.0f) A = 1.0f;
```

The accumulator A must be stored in non-volatile memory at a rate compatible with the allowed number of write cycles.

12 Suggested parameters

Parameter	Value	Notes
E_a	0.5 eV	from qualification or field data
T_{ref}	100 °C	reference junction temperature
L_0	1×10^5 h	expected life at T_{ref}, I_N, $\tau = 2$, $N = 0$
m	2	continuous load profile
$T_{w,th}$	40 °C	inlet water threshold
I_{th}	$0.8\, I_N$	current threshold
β	1	rate doubling in the critical regime
γ	0.05	penalty per intervention
ΔT_w	2 K	thermal threshold regularisation
ΔI	$0.05\, I_N$	current threshold regularisation

Table 2: First-attempt parameter set.

13 Numerical example

Conditions: $I = 0.9\, I_N$, $T_w = 45\,^\circ$C, $T_j = 125\,^\circ$C, $N = 2$, type 2 module.

$$AF_I = 0.9^2 = 0.81 \tag{13}$$

$$AF_{Arr} = \exp\!\left[\frac{0.5}{8.617\times10^{-5}}\left(\frac{1}{373.15}-\frac{1}{398.15}\right)\right] = e^{0.976} = 2.66 \tag{14}$$

$$AF_{th} = 1 + 1\cdot 1 = 2 \quad (T_w > T_{w,th},\ I > I_{th}) \tag{15}$$

$$AF_N = 1 + 0.05\cdot 2 = 1.10 \tag{16}$$

$$\frac{dA}{dt} = \frac{1}{1\times10^{5}}\cdot 0.81\cdot 2.66\cdot 2\cdot 1.10 = 4.73\times10^{-5}/\mathrm{h} \ \Rightarrow\ L \approx 21\,100\,\mathrm{h} \approx 2.4\,\mathrm{yr}. \tag{17}$$

With a type 1 module the rate doubles and the expected life drops to about 10 600 h. Removing the joint threshold condition (for instance by bringing T_w below 40 °C) restores an expected life of about 42 200 h.

14 Model limitations

- The model is linear cumulative (Miner's rule [9]): it does not represent interaction between degradation mechanisms nor irreversible threshold effects.
- The Arrhenius law applied to mean T_j alone does not capture damage from fast thermal cycling; for strongly cyclic mission profiles it must be complemented by rainflow counting [12, 13] on ΔT_j with Coffin–Manson curves [10, 11].
- The coefficients k_τ, β and γ must be identified from field or accelerated test data; the ratio $k_1/k_2 = 2$ is assumed here as a specification input.
- Accuracy depends directly on the quality of the T_j estimate: a 10 K error produces, with $E_a = 0.5\,\mathrm{eV}$, an error of about 30 % on the rate.

References

[1] M. Ciappa, "Selected failure mechanisms of modern power modules," *Microelectronics Reliability*, vol. 42, no. 4–5, pp. 653–667, 2002.

[2] J. Lutz, H. Schlangenotto, U. Scheuermann, R. De Doncker, *Semiconductor Power Devices: Physics, Characteristics, Reliability*, 2nd ed., Springer, 2018.

[3] H. Wang, M. Liserre, F. Blaabjerg, "Toward reliable power electronics: Challenges, design tools, and opportunities," *IEEE Industrial Electronics Magazine*, vol. 7, no. 2, pp. 17–26, 2013.

[4] K. Ma, H. Wang, F. Blaabjerg, "New approaches to reliability assessment: Using physics-of-failure for prediction and design in power electronics systems," *IEEE Power Electronics Magazine*, vol. 3, no. 4, pp. 28–41, 2016.

[5] M. Held, P. Jacob, G. Nicoletti, P. Scacco, M.-H. Poech, "Fast power cycling test of IGBT modules in traction application," in *Proc. Int. Conf. Power Electronics and Drive Systems (PEDS)*, Singapore, 1997, pp. 425–430. (LESIT model)

[6] R. Bayerer, T. Herrmann, T. Licht, J. Lutz, M. Feller, "Model for power cycling lifetime of IGBT modules — various factors influencing lifetime," in *Proc. 5th Int. Conf. Integrated Power Electronics Systems (CIPS)*, Nuremberg, 2008, pp. 1–6.

[7] U. Scheuermann, R. Schmidt, "A new lifetime model for advanced power modules with sintered chips and optimized Al wire bonds," in *Proc. PCIM Europe*, Nuremberg, 2013, pp. 810–817.

[8] Y. Avenas, L. Dupont, Z. Khatir, "Temperature measurement of power semiconductor devices by thermo-sensitive electrical parameters — A review," *IEEE Trans. Power Electronics*, vol. 27, no. 6, pp. 3081–3092, 2012.

[9] M. A. Miner, "Cumulative damage in fatigue," *Journal of Applied Mechanics*, vol. 12, no. 3, pp. A159–A164, 1945.

[10] L. F. Coffin, "A study of the effects of cyclic thermal stresses on a ductile metal," *Transactions of the ASME*, vol. 76, pp. 931–950, 1954.

[11] S. S. Manson, "Behavior of materials under conditions of thermal stress," NACA Technical Note 2933, 1953.

[12] S. D. Downing, D. F. Socie, "Simple rainflow counting algorithms," *International Journal of Fatigue*, vol. 4, no. 1, pp. 31–40, 1982.

[13] ASTM E1049-85, *Standard Practices for Cycle Counting in Fatigue Analysis*, ASTM International.

[14] JEDEC Solid State Technology Association, *JEP122: Failure Mechanisms and Models for Semiconductor Devices.*

[15] IEC 60747-9, *Semiconductor devices — Discrete devices — Part 9: Insulated-gate bipolar transistors (IGBTs).*